

Endogenous Population Scale in the Existing Housing–Fertility Model

Research update

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Abstract

The paper’s lifecycle model determines fertility, tenure, and housing choices for a stationary population. This update adds one feedback: births become future entrant households, and those households add to housing demand. A two-generation model shows how fertility and housing clearing jointly determine long-run housing cost and population. We add this demographic closure around the sequential-fertility model used in the August presentation, without changing the household problem. In units consistent with the model’s 3+ fertility category, the current estimate does not reach replacement over the audited housing-cost range after the fertility decline. This is a conditional model result, not a structural rejection: the causal response of fertility to housing costs is not yet identified.

The numerical exercise indexes the old stationary benchmark to 2007 and phases in a reduced-form fertility-preference decline through 2023. It makes the twenty-year delay from births to adult entry explicit. It does not reproduce the contemporaneous rise in U.S. population or housing costs, nor the movement in the price–rent ratio. A fixed outside entrant flow anchors population in levels; it is an accounting assumption, not a model of migration. The reported first sixty years are therefore a transition experiment between stationary benchmarks, not a fitted history or a welfare calculation.

1 The mechanism

Children raise the demand for space at the same ages when households have low liquid wealth and limited access to ownership. Large rental units are scarce, and buying a family-sized home requires a down payment. Older households, by contrast, own much of the large-housing stock and may retain it after their children leave because moving is costly and housing is part of the estate they plan to leave. The allocation of housing across generations can therefore raise the effective cost of a child even when the aggregate stock is large.

The feedback has two directions. First, a higher housing cost or a tighter access constraint reduces fertility. Second, children eventually form the next generation of households, so fertility changes future housing demand. The first direction is a household mechanism; the second is a demographic equilibrium condition. Combining them changes both the transition and the long-run incidence of housing policy.

The existing model’s intergenerational channel should be stated precisely. Children change their parents’ housing needs, while young and old households compete for the housing stock and face different tenure constraints. The bequest motive also affects old-age saving and housing retention. Estates are not, however, assigned to particular descendants: new entrants draw from a fixed wealth distribution. The present extension adds cohort-size accounting, not an inheritance-transfer mechanism.

These links have close precedents. Demographic effects on housing demand go back at least to Mankiw and Weil (1989), and recent evidence relates birth-cohort size to housing prices through entry into and exit from homeownership (Francke and Korevaar, 2025). Housing prices and mortgage access also affect childbearing (Lovenheim and Mumford, 2013; Dettling and Kearney, 2014; Hacamo, 2021). Malmberg (2012) studies the two-way relation between cohort housing pressure and fertility, while Lee and Suh (2025) analyze a low-fertility housing transition. Our question is narrower: how does the incidence of

housing policy change when tenure constraints and old-age retention affect fertility, and births in turn determine the number of future households?

The current paper already writes a stationary entry condition, while its household computations describe a population normalized to unit size. Let S be the number of adult households in the market. For any housing cost p , the household model tells us four things: the distribution of households, housing demand per household $\bar{h}(p)$, the number of new households required to replace those that exit $E(p)$, and the number produced by children born inside the model $B(p)$. The objects $E(p)$ and $B(p)$ are flows per adult household and model period.

To anchor population in levels, let M be a fixed number of households arriving from outside in each period and let ρ be the fraction of locally born households that remain in the market. This is a transparent accounting closure, not an equilibrium model of migration. A stationary population and a clearing housing market then require

$$SE(p) = M + \rho SB(p), \quad S\bar{h}(p) = H^s(p). \quad (1)$$

The first equation says that retained local children and the fixed outside flow together replace exiting households. The second says that total housing demand equals housing supply. These two equations determine population S and the housing cost p without changing the household problem.

For a stationary comparison, this is the entire population extension. A transition additionally requires the age distribution and the children who have already been born but have not yet formed households. Those objects are introduced below. Outside and locally born entrants are initially assigned the same entry-state distribution as in the current model; allowing their wealth or income distributions to differ is a later extension.

The next section presents the same mechanism with young and old households. Section 3 then maps it into the current paper.

2 Two-generation theory

2.1 Environment

Young households. At date t , there are Y_t young households. Each has income y and chooses consumption c , housing for the adults h , and the number of children n . Preferences are

$$u(c, h, n) = c + \alpha \log h + \theta \log n, \quad \alpha > 0, \quad \theta > 0. \quad (2)$$

The parameter α governs the demand for adult housing and θ governs the demand for children. Quasilinearity makes both demands available in closed form. We assume $y > \alpha + \theta$, so consumption is positive at the interior solution.

Child costs and young access. Each child requires $q \geq 0$ units of other goods and $a > 0$ units of housing. One unit of housing costs p . Obtaining family space also entails an access cost $\omega \geq 0$, which represents a small rental market, a down-payment constraint, or another tenure friction. The cost of a child is therefore $q + a(p + \omega)$.

Old households. There are O_t old households. They no longer choose fertility, and their housing demand is

$$h_O(p) = \frac{\bar{h}_O}{p}, \quad \bar{h}_O > 0. \quad (3)$$

The parameter $\bar{h}_O$ summarizes old households' desire to retain housing. A policy that encourages downsizing reduces $\bar{h}_O$. In the full model, housing retention follows from moving costs and warm-glow utility from estate wealth rather than from this demand schedule. The estate is not transferred to a particular model descendant.

Housing. Aggregate housing services are supplied according to

$$H^s(p) = \bar{H}p^\varepsilon, \quad \bar{H} > 0, \quad \varepsilon \geq 0. \quad (4)$$

The fixed-stock case is $\varepsilon = 0$.

Demography. The number $\bar{n} > 0$ is replacement fertility. After survival and household formation, $\bar{n}$ children produce one new young household.

In this static section, p is the current cost of one unit of housing services. The lifecycle model distinguishes that current rent, denoted r_t , from the asset price P_t that capitalizes future rents. They move proportionally only under a constant user cost and a stationary price path.

2.2 The young household

Taking p as given, a young household solves

$$\max_{c, h, n} c + \alpha \log h + \theta \log n \quad \text{subject to} \quad c + ph + [q + a(p + \omega)]n = y. \quad (5)$$

At an interior solution, the household chooses

$$n(p) = \frac{\theta}{q + a(p + \omega)}, \quad h_Y(p) = \frac{\alpha}{p} + an(p). \quad (6)$$

Thus a higher housing cost or a tighter access constraint reduces fertility. Young housing demand is adult housing, α/p , plus the space required by children, $an(p)$.

This is a stripped-down version of the household mechanism already in the paper. There, the other-goods cost of children is χ , the space required per child is κ , and the relevant housing cost depends on tenure and on whether the household is constrained. In the notation of the current model, the simple expression $q + a(p + \omega)$ corresponds to $\chi + \kappa(r^m + \zeta_i^m)$: ω summarizes the shadow cost of limited housing access. The simple model removes heterogeneity, saving, and the rent-versus-own decision only to make the demographic feedback transparent.

2.3 Equilibrium and the steady state

Housing clearing adds young and old demand:

$$\bar{H}p_t^\varepsilon = Y_t h_Y(p_t) + O_t h_O(p_t). \quad (7)$$

For inherited cohort masses, the left side rises and the right side falls with p_t , so the clearing price is unique.

Children become the next young cohort, while the current young become old:

$$Y_{t+1} = \frac{n(p_t)}{\bar{n}} Y_t, \quad O_{t+1} = Y_t. \quad (8)$$

The ratio $n(p_t)/\bar{n}$ is the number of next-period young households produced by each current young household.

A steady state has a constant housing cost and constant positive cohort masses, so $n(p^*) = \bar{n}$ and $Y^* = O^*$.

Proposition 1 (Steady state). *A unique positive steady state exists if and only if $\theta > \bar{n}(q + a\omega)$. It is*

$$p^* = \frac{\theta/\bar{n} - q}{a} - \omega, \quad Y^* = O^* = \frac{\bar{H}(p^*)^{\varepsilon+1}}{\alpha + \bar{h}_O + a\bar{n}p^*}. \quad (9)$$

A lower value of children θ or a higher access cost ω lowers both the steady-state housing cost and population. A higher $\bar{h}_O$ reduces population but does not change the steady-state housing cost. A proportional increase in housing supply $\bar{H}$ raises population in the same proportion and also leaves the steady-state housing cost unchanged.

The cohort law first determines the housing cost consistent with replacement, $n(p^*) = \bar{n}$. Housing clearing then determines how many households can live in the market at that cost. This is why a permanent increase in housing supply raises population rather than lowering the steady-state housing cost.

Even if market housing were free, a child would still cost $q + a\omega$. Hence cheaper housing cannot restore replacement when $\theta \leq \bar{n}(q + a\omega)$.

2.4 Comparative statics

Suppose the economy begins in a steady state and a permanent reform arrives after Y_t and O_t have been inherited. The impact and steady-state effects are:

	Impact p_t	Impact n_t	Long-run p^*	Long-run n^*	Long-run population
Lower access cost, $\omega \downarrow$	+	+	+	0	+
Lower old housing demand, $\bar{h}_O \downarrow$	−	+	0	0	+
Greater housing supply, $\bar{H} \uparrow$	−	+	0	0	+

Lowering ω makes family housing easier to obtain. Fertility and housing demand rise immediately, so part of the reform appears as a higher market price. Lower old housing demand and greater supply instead reduce the price on impact. In the new steady state, fertility is again $\bar{n}$. For the latter two reforms, the housing cost returns to its original level and the additional housing supports a larger population.

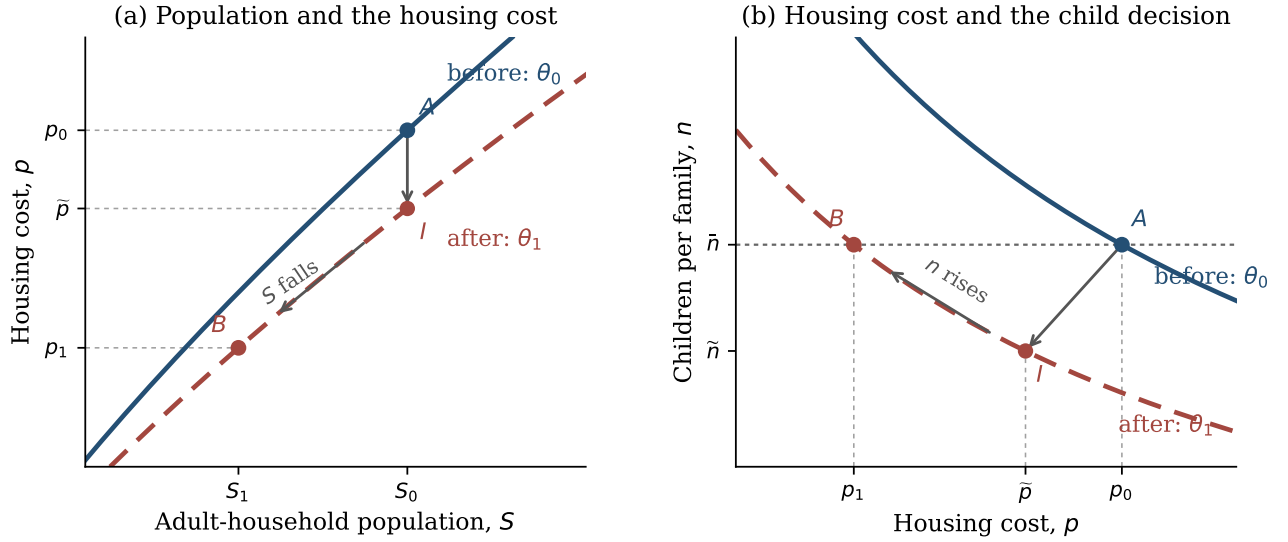


Figure 1: The child decision determines the steady-state housing cost through $n(p^*) = \bar{n}$. Housing clearing then determines population at that cost. A fall in θ initially puts fertility below replacement; as population falls, housing becomes cheaper and partly offsets the initial decline. The solid blue housing-clearing schedule applies before the fall in θ and the dashed red schedule after it; A, I, and B denote the initial steady state, impact allocation, and new steady state.

2.5 Demographic momentum

Let $\mu_t = O_t/Y_t$ be the number of old households per young household, and let $m_t = n(p_t)/\bar{n}$ be the number of next-period young households produced by each current young household. When fertility is below replacement, $m_t < 1$, total household population still rises if

$$\mu_t < m_t < 1. \quad (10)$$

For example, 100 young and 70 old households become 80 young and 100 old households when $m_t = 0.8$. Population rises from 170 to 180 even though the young cohort is not replacing itself.

If preferences and housing supply do not change again between t and $t + 1$, the housing cost also rises when

$$(1 - \mu_t)h_O(p_t) > (1 - m_t)h_Y(p_t). \quad (11)$$

The smaller young cohort reduces demand by $(1 - m_t)Y_th_Y(p_t)$. At the same time, the current young cohort replaces a smaller old cohort and raises old demand by $(1 - \mu_t)Y_th_O(p_t)$. The price rises when the second change is larger.

Starting from an exact steady state, $\mu_t = 1$. In this closed two-cohort model, a fertility decline lowers the housing cost on impact and lowers population in the next period. A period in which fertility is below replacement while population and housing costs rise therefore requires an inherited cohort imbalance, produced by earlier fertility, immigration, or both.

On impact, the fertility decline lowers the housing cost relative to the no-shock allocation. Demographic momentum can nevertheless make the realized cost rise over time. The sign comparison above does not by itself order the entire shocked and no-shock price paths.

2.6 Intergenerational policy

Lowering ω directly improves young households' access to family space. Lowering $\bar{h}_O$ instead releases housing from old households and reduces the market price during the transition. Both policies raise fertility, but through different margins. A general expansion of supply also lowers the transition price, without changing which generation initially releases the housing.

With a fixed adult population, reallocating a home toward a constrained young household affects current tenure and fertility. With the cohort law in equation (8), the resulting birth also changes the number of future entrants. That second effect is lost when population is renormalized after every solution.

3 Quantitative implementation

3.1 Minimal wrapper around the sequential model

The quantitative model is the one-market sequential-fertility specification used in the August presentation. Households make age-dependent first- and later-birth decisions, children ever born are distinct from children currently at home, and the model retains income risk, saving, tenure choice, the owner ladder, and a warm-glow bequest motive. The population closure does not add a new household state or alter these decisions. It also does not make bequests flow to descendants; entrant wealth remains exogenous. At a candidate stationary housing cost P , the solver returns the adult entry flow $E(P)$, effective locally born entrant households $B(P)$, and housing demand per adult household $D(P)$. In the stationary solver, rent and the house asset value move proportionally through the maintained user-cost equation. For a fixed outside entrant flow M and local retention ρ , stationary adult-household population is

$$S(P) = \frac{M}{E(P) - \rho B(P)}. \quad (12)$$

The denominator is the entry shortfall left after retained local children. The fixed-inflow stationary population is finite when this denominator is positive. Housing clearing then chooses P from

$$S(P)D(P) = H^s(P). \quad (13)$$

Thus the stationary extension is an outer calculation around the existing sequential model. It does not require a new household problem.

For comparison, a positive closed steady state would require both replacement and housing clearing:

$$B(P^*) = E(P^*), \quad S^* = \frac{H^s(P^*)}{D(P^*)}. \quad (14)$$

The first condition need not have a solution at the estimated household parameters. This is a substantive restriction, not a numerical normalization.

3.2 Replacement accounting and the closed root

The calculation holds the working parameter estimate fixed. That estimate is not yet a final calibration: it misses childlessness, mean occupied rooms, and old-age wealth dispersion, and its first-birth housing target remains under empirical review.

One accounting correction is essential. The household state space records parities 0, 1, 2, 3+, so its raw child pipeline carries at most three explicit children. The reported completed-fertility moment instead assigns the 3+ state its observed mean, 3.602. Comparing the raw maturation flow to required entry therefore mixes child units. For stationary renewal, we assign the extra 0.602 children to each flow entering the 3+ state and give them the same maturation yield as the explicit children. This changes no household choice.

Under the fiscally balanced 1 percent property-tax convention, required entrants are $E_0 = 0.061733$ per adult household. The raw three-child state delivers 0.052929 mature entrants; the top-bin-consistent flow is $B_0 = 0.057798$. The replacement ratio relevant for the closure is therefore

$$\frac{B_0}{E_0} = 0.936249, \quad (15)$$

not 0.857387. The latter is retained only as a raw-state diagnostic.

The estimated fertility response has the expected sign but is small. Reducing the housing cost to one half of one percent of its baseline value raises the consistent $B(P)/E(P)$ ratio only to 0.957554. The schedule is monotone and does not cross one over the audited range from 0.5 to 300 percent of the baseline cost. Equation (14) therefore has no root on that range at the current parameters. By contrast, the old preference benchmark chosen to deliver completed fertility of 2.12 has $B/E = 1.003$ in the same units, so treating it as the initial stationary benchmark is internally coherent to numerical accuracy.

This no-root result is conditional on an incompletely identified slope. The fertility levels and timing used in the current calibration mainly determine the preference intercepts at one housing cost; they do not identify the causal change in fertility when housing costs move. A slope sensitivity reported below therefore reanchors the old and post-decline fertility levels before changing the logit scales. It is useful for feasibility, but it is not a replacement for a housing-cost-to-fertility moment.

Figure 2 shows the two stationary objects. The left panel is the replacement condition; the right panel records the population scale allowed by housing clearing at each cost. The solid line uses the top-bin-consistent child units; the dashed line shows why the raw three-child state cannot be used for this test. The present estimate supports a fixed-inflow endpoint over the displayed range, not a closed one.

A deliberately uncalibrated sensitivity confirms that the model class can produce the feedback in the simple theory. Multiplying both fertility-logit scales by 0.10 makes choices much more responsive, while

Stationary schedules in the sequential model

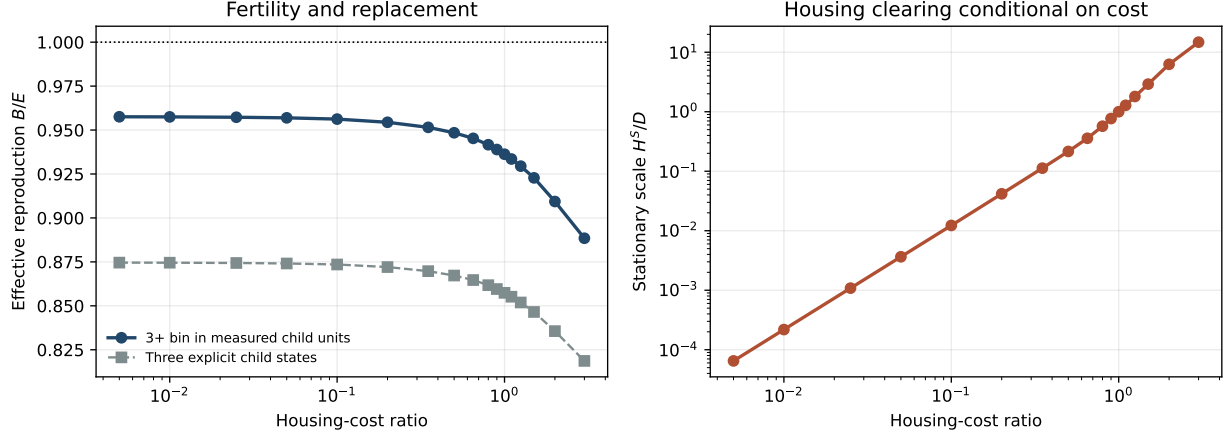


Figure 2: Stationary schedules in the sequential model. The left panel plots effective locally born entrant households relative to required entrants. The solid line counts the 3+ state in the same units as measured completed fertility; the dashed line counts only three explicit children. The right panel plots the scale permitted by the maintained housing-supply schedule, conditional on housing cost. The working estimate remains below replacement throughout the displayed range.

the preference intercepts are reanchored to the same old and post-decline fertility levels. Replacement then occurs at a housing cost equal to 0.078 of its old level. With a multiplier of 0.25, no root appears; with a multiplier of 0.05, a root appears at 0.499 of the old housing cost. These are feasibility checks, not alternative estimates: they show both that the feedback can operate and that its quantitative strength is presently undisciplined.

3.3 How the closure changes the paper’s policy experiments

The standing fiscally balanced policy cases are also run under the sequential model. The reference is a 1 percent annual property tax on occupied housing, rebated equally. The first counterfactual raises the rate to 2 percent. The second also provides a purchase grant of 0.4 model income units to a dependent-child renter who buys a home with at least six rooms. Residual tax revenue is rebated equally in both counterfactuals. The parameter estimate is fixed, so these are mechanism calculations rather than final policy estimates.

The diagnostic assigns 16.9 percent of baseline entrants to outside origin. This number comes from across-market mobility and is most naturally read as a local-market anchor, not as national immigration. Local retention is chosen to reproduce the reference population exactly; the top-bin-consistent accounting implies $\rho = 0.8876$. That retention rate and the outside flow are then held fixed. Total births combine births per adult household with the change in adult-household population.

Table 1: Policy effects under the fixed-inflow stationary closure

	ΔP	ΔS	Δ births/adult	Δ total births
Rebated 2% property tax	-17.47	+1.91	+0.38	+2.29
Rebated 2% tax + purchase grant	-17.17	+3.84	+0.75	+4.61

Notes: Percent changes from the fiscally balanced 1 percent property-tax reference. Population is adult-household mass; births use the top-bin-consistent child units defined above.

At the current parameter values, the housing-cost response remains the main effect, as in the fixed-population exercise. The closure adds a second margin: lower housing costs raise births per household and, after children form households, the stationary market contains more households. This demographic magnitude is conditional on the model’s housing-cost-to-fertility slope, which the current target system

does not identify causally. The rows therefore show model mechanics under one slope and one provisional outside-flow anchor; they are not empirical pronatal-policy estimates, transition paths, or welfare results.

3.4 A transition indexed to 2007

The calendar wrapper carries the full adult distribution over age, liquid wealth, income, tenure, parity, and children at home. Its one-period operator reproduces the stationary distribution and the top-bin-consistent birth and entrant flows with errors below 10^{-12} . Population mass is never renormalized.

The household child state is deliberately not used as the demographic clock. Although it tracks children’s departure independently, its geometric transition is designed for housing demand rather than exact birth-vintage timing. Population renewal therefore carries a separate aggregate queue: births affect parents’ housing choices immediately, but affect adult entry twenty years later. The model’s final parity state is 3+. For renewal accounting, families entering that state receive the same mean number of children, 3.602, used in the reported completed-fertility statistic. The extra children are assigned the same maturation rate and timing as the third birth. This accounting bridge leaves household choices unchanged, but it is an approximation to the timing of fourth and later births. During the preference transition, the mature flow from this dated queue differs from the household child-state clock by as much as 13.0 percent. That gap is a transparent timing choice, not a market-clearing error.

The diagnostic asks a deliberately narrow question. We index the old stationary benchmark to 2007; we do not claim that the U.S. economy was then in steady state. The child-preference shifter moves linearly from 0.1062 in 2007 to -0.2419 in 2023. At the initial housing cost, these values imply model top-coded completed-fertility statistics of 2.120 and 1.616. The latter is not the new stationary value because housing costs subsequently adjust. The corresponding U.S. observations are period total fertility rates, not completed fertility in a stationary cross-section. The preference path is therefore a reduced-form normalization, not an estimated structural shock.

The outside flow remains 16.9 percent of initial entry and local retention is fixed at the value that balances the old benchmark, $\rho = 0.8286$. Both are held fixed rather than determined by migration choices. Table 2 reports what the preference change alone generates through 2023. Population cannot yet respond because the affected births have not reached adult entry. Housing demand responds immediately, but only modestly. The model therefore predicts essentially flat population and slightly lower housing costs, while U.S. population and real housing costs rose.

Table 2: The 2007–2023 path: data and model

	Population	Real house price	Real rent	Price/rent
U.S. data	+11.73	+15.62	+15.88	−0.23
Fixed housing stock	0.00	−0.71	−0.71	0.00
Static elastic supply	0.00	−0.15	−0.15	0.00

Notes: Percent changes from annual 2007 averages. House prices and rents are deflated by the all-items CPI. The model population is adult-household mass; the U.S. series is total persons. The other U.S. series are the Case-Shiller national index, rent of primary residence CPI, and total CPI, downloaded from FRED.

Figure 3 makes the miss visible rather than absorbing it in another shifter. The maintained stationary relation $r_t = (R_b - 1 + \delta_H + \tau_t^p)P_t$ fixes the model price-rent ratio. It therefore cannot reproduce the sharp fall in the U.S. ratio after 2007 and its subsequent recovery, even though the ratio happens to be close to its 2007 value again by 2023.

Within this model, holding other aggregate forces fixed, the result rules out using the fertility trend alone to explain the aggregate post-2007 housing boom. The paper can instead make a relative claim: the fertility decline lowers housing demand and eventually population relative to a no-decline path.

A mechanical existence check confirms that the sign conflict is not built into the model. Multiplying the four already-born cohorts in the future-entrant pipeline by 1.572, chosen solely to match the 2007–2023 growth of U.S. total population, makes the fixed-stock housing-cost index rise by 16.97 percent; with

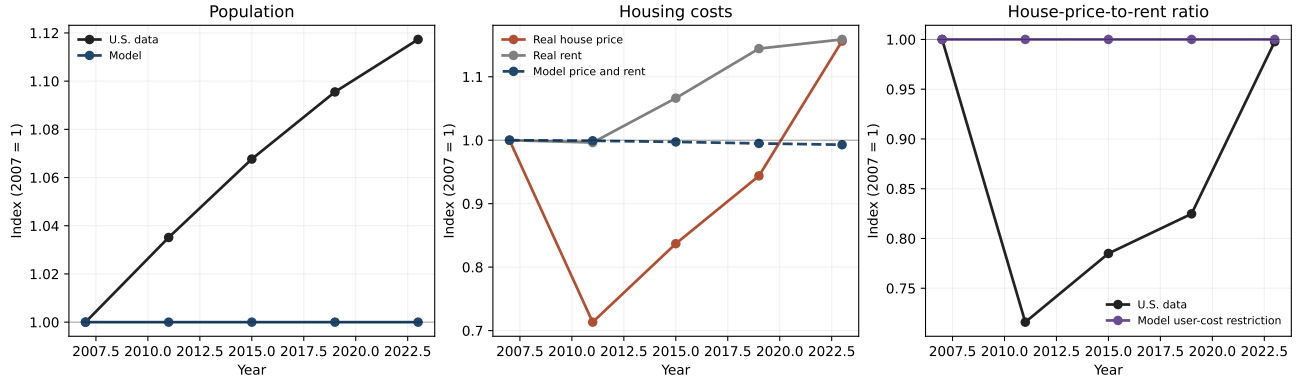


Figure 3: Historical comparison for the fixed-stock transition. The old stationary benchmark is indexed to 2007 and only the fertility-preference shifter changes. Real house prices and rents are nominal indexes divided by the all-items CPI. The model’s price and rent indexes coincide because its stationary user-cost coefficient is constant. Population is total persons in the data and adult households in the model.

elastic supply it rises by 3.77 percent. This is not yet a defensible demographic initialization. U.S. births rose only 7.36 percent between 1987 and 2003 (Martin et al., 2005), while the number of households headed by someone under 25 fell 7.90 percent between 2007 and 2023. Total households rose 13.29 percent over those dates (U.S. Census Bureau, 2025). A coarse reweighting of the model’s initial age distribution to the Census’s 2007 householder-age shares produces only 3.04 percent growth by 2023 at the steady-state entrant flow. The missing object is therefore a dated household-formation and migration residual, not simply a larger birth queue.

The long-run housing-supply convention matters. With the initial stock fixed forever, the post-shock population demands less housing than that stock throughout the audited grid, which extends down to $P = 10^{-4}$. A usable positive-price endpoint therefore requires vacancies, depreciation, or scrappage. The simpler alternative is the sequence of static equilibria used by Coeurdacier et al. (2023): at each date a constant-elastic housing-supply schedule clears, and the demographic distribution is then advanced. Using the maintained supply elasticity of 1.75, the limiting fixed-inflow stationary point is:

$$\frac{S^{\text{new}}}{S^{\text{old}}} = 0.4697, \quad \frac{P^{\text{new}}}{P^{\text{old}}} = 0.7375, \quad \text{model completed fertility}^{\text{new}} = 1.633. \quad (16)$$

Cheaper housing partly offsets the preference decline, but local reproduction remains below replacement and the fixed outside flow supports a smaller city. The static schedule is a tractable benchmark, not a literal law of motion for a durable stock: it lets supply contract as readily as it expands. The smallest realistic extension would treat that curve as a desired stock and allow slow downward adjustment through vacancies, depreciation, and demolition. That asymmetric stock adjustment has not yet been estimated or implemented.

To reconnect the transition to the paper’s tenure mechanism, a companion path raises the financed share to 0.95 for every dependent-child household from 2023 onward. This is an eighteen-year expected child-spell policy, not a one-period new-parent intervention. Sixty years after the initial date, it raises dependent-child ownership by 4.13 percentage points and aggregate ownership by 1.53 points relative to the no-policy path. Adult-household population is only 0.002 percent higher and housing cost 0.022 percent higher. At the current slope, the access reform therefore has a visible tenure effect but essentially no demographic amplification over this horizon.

After sixty years, adult-household mass is 0.8852 and the housing-cost index is 0.9517. Even after 236 years they remain at 0.5051 and 0.7599, above their limiting values, so convergence is slow. The calculation follows Oswald’s tractable timing but not a forward-looking asset-price equilibrium: households treat the current price and current preference shifter as permanent at each date. It can therefore describe demographic and housing-demand accounting, but not welfare or expected capital gains.

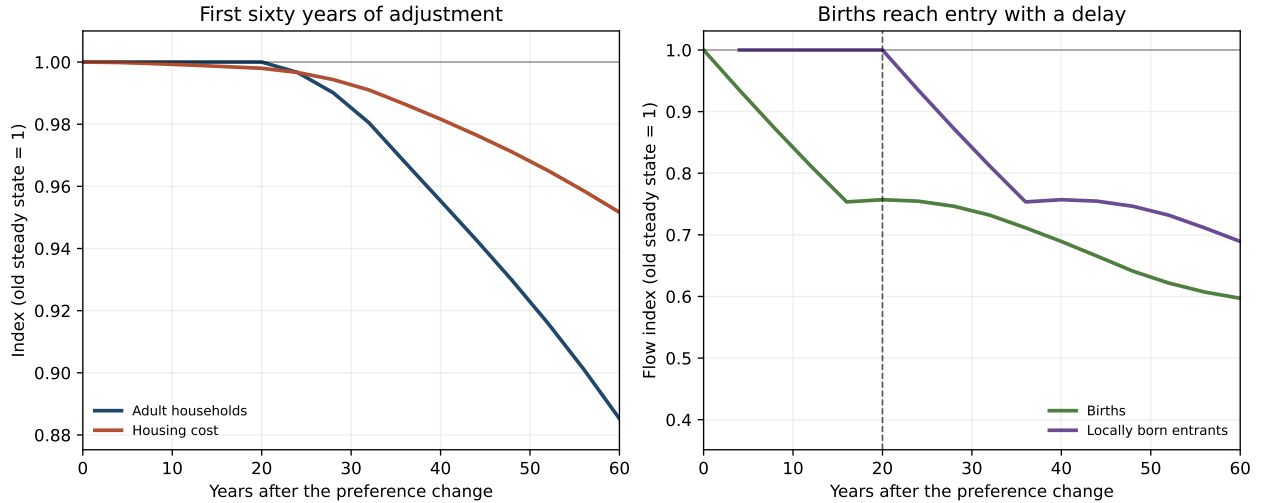


Figure 4: Movement under static elastic housing supply. Births fall while locally born entry remains at its old flow until the birth-vintage queue turns over. Adult-household population is therefore unchanged for twenty years and then declines. The figure shows the first sixty years; the dashed vertical line marks the twenty-year entry delay, and equation (16) reports the limiting fixed-inflow levels.

3.5 Interpretation

The minimal paper should not call this a fitted U.S. transition. It can index an old stationary benchmark to 2007, interpret the preference path as a reduced-form fertility trend, and compare housing policies from the 2023 point along that path. The aggregate data are then descriptive benchmarks, not calibration targets. The historical miss is reported explicitly.

A literal U.S. transition would require two further objects. First, the model needs the inherited age and household-formation distribution that generated population growth after 2007. Simply inserting the ACS head-age distribution is not valid because the model treats age 18 as household entry. Second, households must anticipate rents and resale prices. That requires the calendar-time Bellman problem and dynamic user-cost equation above, rather than a sequence of stationary household problems.

Finally, the long-run strength of the feedback depends on the causal response of fertility to housing costs and access. Fertility levels at one price do not identify that slope. The current no-root result should therefore be revisited after the fertility and first-birth housing targets are repaired and the model is refit. This is the closed-population replacement root discussed above; it is separate from the fixed-stock vacancy corner in the transition.

4 Conclusion

The proposed change is narrow. A stationary outer loop uses the existing model's entry, child-maturation, and housing-demand objects to determine adult- household population and housing cost jointly; a birth-vintage queue adds the transition. The sequential household problem is unchanged. The paper's intergenerational content remains the allocation of housing across cohorts and parents' housing response to children, not a transfer of estates to model descendants.

In corrected child units, the working estimate has $B/E = 0.936$ and no closed root on the audited housing-cost range. Because the housing-cost-to-fertility slope is not identified, this is a conditional result rather than a rejection of the mechanism. The old preference benchmark is nevertheless internally close to replacement, with $B/E = 1.003$. After the reduced-form fertility decline, population is flat through 2023, then falls with a twenty-year delay; after sixty years household mass is 0.885 and housing cost is 0.952. Under static elastic supply the limiting fixed-inflow levels are 0.470 and 0.737.

This path does not explain the post-2007 U.S. housing boom or the price-rent cycle. It is a coherent

cohort-accounting benchmark for asking who owns the housing stock and when policy effects reach future cohorts. At the current slope, the dependent-child credit reform meaningfully raises ownership but has almost no demographic effect. The next empirical priority is therefore the causal fertility response to housing costs and access; a fitted U.S. transition would additionally need inherited household-formation and migration momentum, asymmetric housing-stock adjustment, and eventually forward-looking asset prices.

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